REAL-TIME EMBEDDED SYSTEM
EXPERIMENT NO. 6

SELF-DRIVING CAR

Name: Lelis, Kathryna T.

Course/Section: CPE161P-4/C1

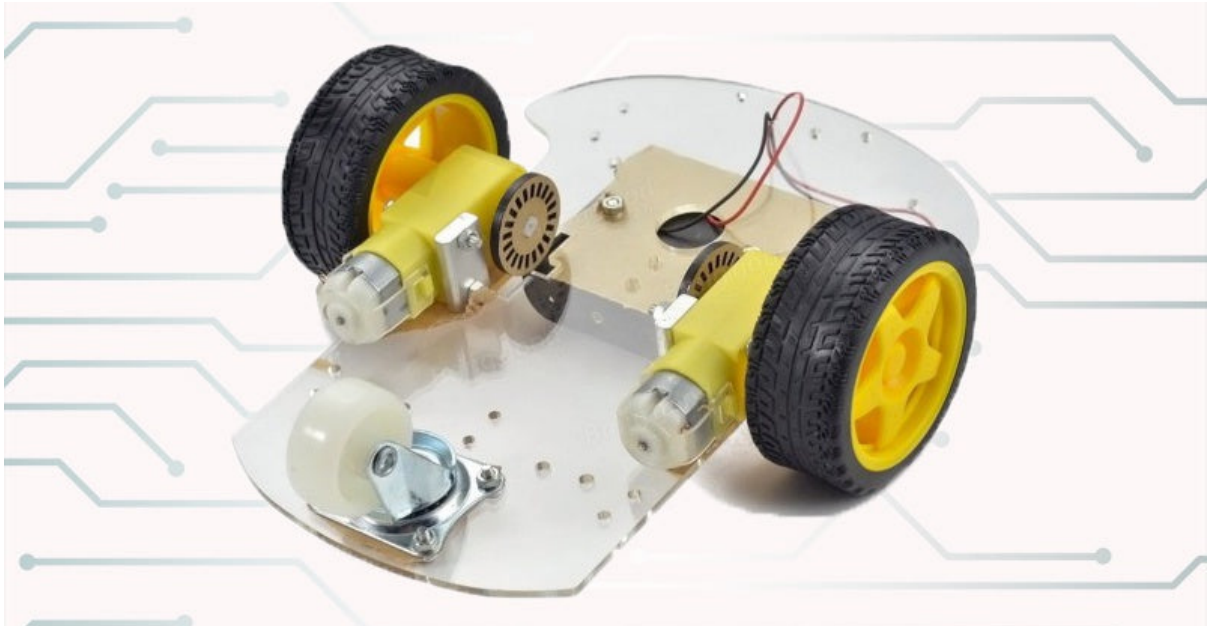
Group No.: INDIVIDUAL

Date of Performance: January 21, 2024
Date of Submission: January 25, 2024

Cyrel O. Manlises, Ph.D.
Instructor

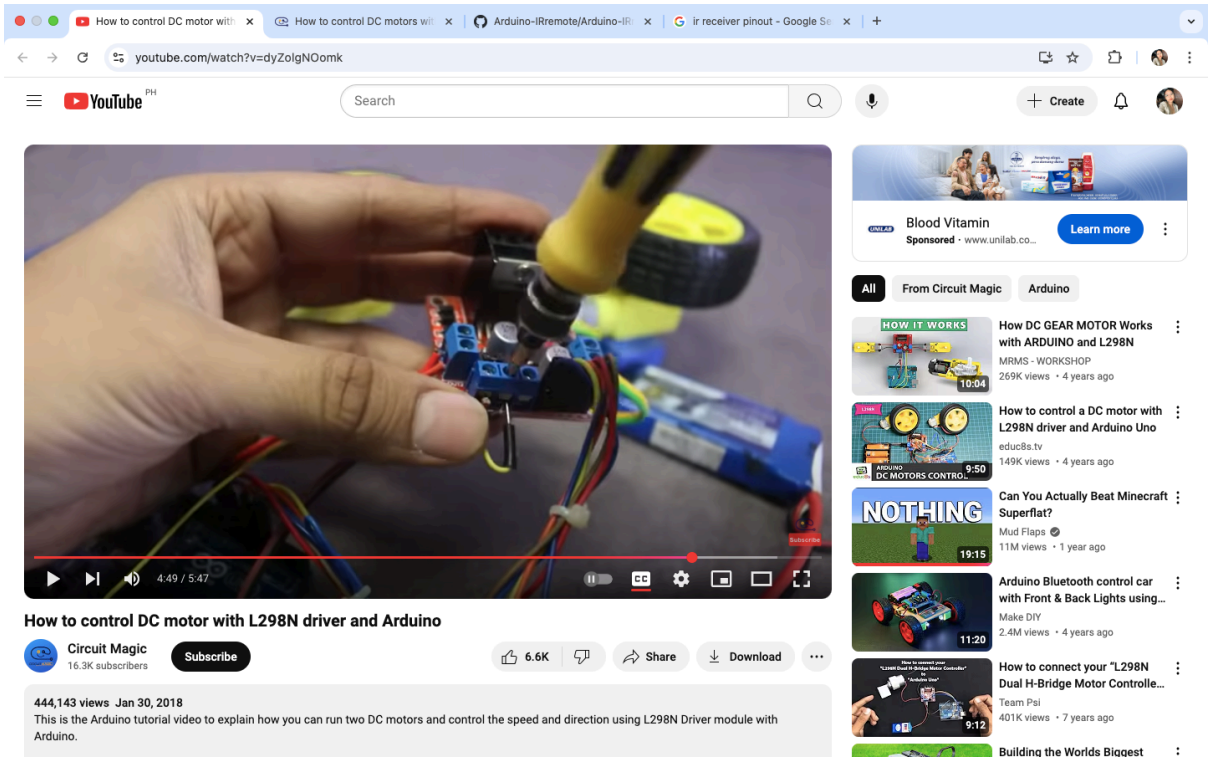
DISCUSSION

Experiment 6 revolves around the idea of a Self-Driving Car or a car itself. For this purpose, I utilized a 2-wheel car chassis kit for easier assembly of components to act as the car's body.



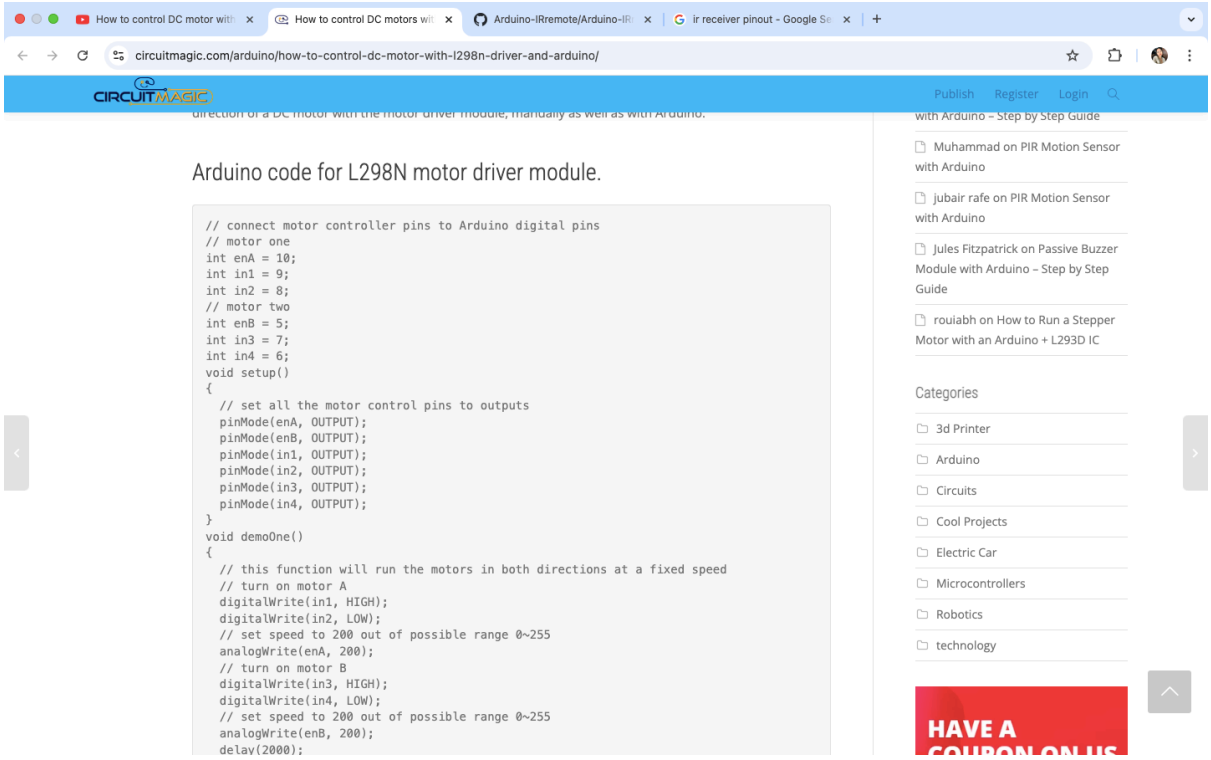
Therefore, since this is the first time I will be able to assemble and use this to connect with the Arduino, I conducted research and watched video tutorials on how to properly assemble and wire the chassis kit to the Arduino Uno board. I discovered that I also need

a DC motor driver which is a L298 DC motor driver commonly use with this car chassis kit.



Through the tutorials, I learned about the directional control patterns and the wiring configuration required to connect the kit to the Arduino effectively. One key detail was ensuring that the ground connections of the motor and the Arduino are commonly connected to the driver for proper

functionality



With this reference code, I combine it with the code of IR remote used in experiment 4. IR remote is just an additional feature so we can play the sequence as well as drive it manually whenever we want. This is to limit the car to bump to objects and to avoid damages.

CONCLUSION

The experiment met successful in showing the basic functionality of a self-driving car: it was capable of moving both forward and backward, thus fully accomplishing the objective of how such systems work. It also allowed the application of theoretical knowledge related to logic circuits, switching theory, and microprocessors to designing and implementing an actual system. A real-time embedded system was implemented by adding an Arduino Uno board, incorporating components like the L298 DC motor driver and IR remote, into the chassis of the car to control movements effectively. The other feature in the IR remote was manual control, which allowed safety and flexibility in that the car can adjust to dodging obstacles that do not cause damage. This

hands-on experiment bridged theoretical knowledge with practical implementation and offered valuable insight into embedded systems and self-driving car technology, all the while consolidating key principles of logic design and microprocessor applications.